

StudyMatch Survey Redesign

Three MSLQ-Informed Versions of the 24-Item Instrument

DS 440 Data Science Capstone — Report 3 Working Draft

The Motivated Strategies for Learning Questionnaire (MSLQ) is an 81-item, validated instrument with 15 subscales measuring how students study, but it was designed to measure solo study behavior — it has no subscale for how someone behaves inside a group. Our collaboration axis is kept unchanged across all three versions below for that reason; every MSLQ item referenced here is paraphrased for group-study context rather than reproduced verbatim from the original instrument.

Each version below applies MSLQ to the other five axes — planning, session_mode, reliability, structure, and intensity — with a different rule for how much of each axis's content comes from MSLQ versus from our original items.

At a glance

Version	Approach
1 — Broad Mix	Each axis draws on whichever MSLQ subscales fit best, mixed freely.
2 — One Subscale, One Axis	Each axis maps to exactly one MSLQ subscale, cleanly.
3 — Recommended Blend	MSLQ where it fills a real gap, original items where MSLQ doesn't apply, hybrid where both matter.

All three keep the same six axes, the same 6-point scale, and 4 items per axis (24 total). Only the sourcing of the non-collaboration items changes.

Version 1 — Broad Mix

Each axis draws on whichever MSLQ subscales fit it best, without restricting to a single subscale. This version optimizes for content fit over methodological cleanliness.

planning

PLANNING	<i>MSLQ: Metacognitive Self-Regulation + Time & Study Environment Management</i>
1.	I plan out what I need to cover before a study session starts.
2.	I keep track of upcoming deadlines well ahead of time.
3.	When I realize my plan for a session isn't working, I adjust it instead of pushing through.
4.	I often start studying without a clear sense of what I want to accomplish. (R)

session_mode

SESSION_MODE	<i>MSLQ: Time & Study Environment Management</i>
5.	I do my best studying in a quiet, distraction-free space.
6.	I get more done when I study in short, focused blocks rather than long stretches.
7.	Background noise and activity around me don't affect how well I focus. (R)
8.	I prefer studying at a set time and place rather than wherever or whenever it fits.

reliability

RELIABILITY	<i>MSLQ: Effort Regulation</i>
9.	I follow through on study commitments even when I'd rather be doing something else.
10.	If material gets boring or difficult, I'm still able to push through it.
11.	I sometimes give up on a task before finishing if it stops being interesting. (R)
12.	I show up prepared to group sessions, having done my part beforehand.

structure

STRUCTURE	<i>MSLQ: Organization + Metacognitive Self-Regulation</i>
13.	I organize my notes and materials in a way I can easily find later.
14.	I break larger assignments into smaller steps before starting.
15.	My study materials tend to be scattered and disorganized. (R)
16.	I go back over material I've already covered to check what actually stuck.

intensity

INTENSITY	<i>MSLQ: Self-Efficacy + Task Value + Goal Orientation (mixed)</i>
17.	I'm confident I can do well in this course if I put in the effort.
18.	I care about actually understanding the material, not just getting a good grade.
19.	I doubt my ability to grasp the harder material in this course. (R)
20.	Getting a good grade matters more to me than learning the material itself. (R)

collaboration

COLLABORATION	<i>Original StudyMatch items — unchanged</i>
21.	I learn material best by explaining it to someone else.
22.	I'd rather listen to someone explain it than explain it myself. (R)
23.	I'm comfortable being the person who answers the group's questions.
24.	I'd rather attempt problems alone first and bring questions to the group. (R)

Kept identical across all three versions. MSLQ is a study-alone instrument and has no items measuring how someone behaves inside a group, so there is nothing to source here.

Version 2 — One Subscale, One Axis

Each axis maps to exactly one MSLQ subscale, using what we judged the four strongest items from that subscale. This version optimizes for a clean, citable one-to-one correspondence between our axes and an established instrument's subscales.

planning

PLANNING	<i>MSLQ: Metacognitive Self-Regulation</i>
1.	I set clear goals for what I want to accomplish before I start studying.
2.	I notice when a study strategy isn't working and try something else.
3.	I ask myself questions about the material to check my own understanding.
4.	I rarely stop to think about whether my approach to studying is actually working. (R)

session_mode

SESSION_MODE	<i>MSLQ: Time & Study Environment Management</i>
5.	I have a regular place where I do most of my studying.
6.	I manage my time well enough to avoid last-minute cramming.
7.	I get easily distracted no matter where I try to study. (R)
8.	I try to study during the times of day when I focus best.

reliability

RELIABILITY	<i>MSLQ: Effort Regulation</i>
9.	Even when material is dry, I make myself keep working through it.
10.	I keep working on something difficult instead of giving up on it.
11.	I often lose motivation partway through a task and don't finish it. (R)
12.	I hold myself to the commitments I make, even outside of a deadline.

structure

STRUCTURE	<i>MSLQ: Organization</i>
13.	I outline or organize material before diving into the details.
14.	I keep my study materials organized so I can find what I need quickly.
15.	My notes tend to be messy and hard to use later. (R)
16.	I make lists of the tasks I need to complete for a course.

intensity

INTENSITY	<i>MSLQ: Self-Efficacy for Learning & Performance</i>
17.	I'm confident I can understand the most complex material in this course.
18.	I believe I can do well in this course if I apply myself.
19.	Compared to others in this course, I don't think I'm very capable. (R)
20.	I expect to do well in this course.

collaboration

COLLABORATION	<i>Original StudyMatch items — unchanged</i>
21.	I learn material best by explaining it to someone else.
22.	I'd rather listen to someone explain it than explain it myself. (R)
23.	I'm comfortable being the person who answers the group's questions.
24.	I'd rather attempt problems alone first and bring questions to the group. (R)

Identical to Version 1 — see note above.

Version 3 — Recommended Blend

This version keeps MSLQ where it measures something our original axis didn't capture well, keeps our original items where MSLQ was never designed to measure that construct, and blends the two where both halves of an axis matter. The sourcing rule behind each decision is stated with the axis.

planning

PLANNING	<i>MSLQ: Metacognitive Self-Regulation</i>
1.	I set clear goals for what I want to accomplish before I start studying.
2.	I notice when a study strategy isn't working and switch to something else.
3.	I check my own understanding as I go, instead of waiting for a quiz to find out.
4.	I rarely stop to think about whether my approach to studying is actually working. (R)

Rationale: our original planning items were thin on metacognitive monitoring. MSLQ's Metacognitive Self-Regulation subscale measures exactly that gap, with no group-specific meaning lost by adopting it wholesale.

session_mode

SESSION_MODE	<i>Original StudyMatch items — kept, not MSLQ</i>
5.	I prefer sessions where we talk through problems out loud together.
6.	I'd rather work through material quietly on my own, even in a group setting. (R)
7.	I like sessions that follow a clear agenda rather than open-ended discussion.
8.	I do better in loosely structured sessions where the conversation can wander. (R)

Rationale: MSLQ's Time & Study Environment Management measures a solo physical setup (quiet room, fixed schedule), not a group session's verbal style or agenda. That's a different construct MSLQ was never built to capture, so we kept the original items rather than force a substitution.

reliability

RELIABILITY	<i>Hybrid: 2 MSLQ Effort Regulation + 2 original items</i>
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9.	Even when material is dry, I make myself keep working through it.
10.	I keep working on something difficult instead of giving up on it.
11.	I show up to group sessions having done what I said I would beforehand.
12.	I've flaked on a study commitment to my group before. (R)

Rationale: MSLQ's Effort Regulation captures internal follow-through on hard material, but says nothing about follow-through toward other people. Since reliability is our highest-weighted axis and flaking is the most common cause of group collapse, we kept both halves rather than pick one.

structure

STRUCTURE	<i>Hybrid: 2 MSLQ Organization + 2 original items</i>
13.	I outline or organize material before diving into the details.
14.	I keep my study materials organized so I can find what I need quickly.
15.	I like group sessions to follow a set structure or checklist.
16.	I'm fine with sessions that don't have a clear plan going in. (R)

Rationale: MSLQ's Organization subscale is about structuring material (outlines, notes). Our original items are about structuring the session itself (agenda, checklist). Both are "structure" but at different levels, so both stay.

intensity

INTENSITY	<i>MSLQ: Self-Efficacy + Task Value</i>
17.	I'm confident I can understand the most complex material in this course.
18.	I believe I can do well in this course if I apply myself.
19.	I care about understanding the material, not just getting a good grade.
20.	Compared to others in this course, I don't think I'm very capable. (R)

Rationale: our original intensity items were the weakest of the original 24 — socially desirable and low-variance on our highest-weighted axis alongside reliability. MSLQ's Self-Efficacy and Task Value subscales are validated to actually spread respondents out here, so we replaced rather than blended.

collaboration

COLLABORATION	<i>Original StudyMatch items — unchanged</i>
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21.	I learn material best by explaining it to someone else.
22.	I'd rather listen to someone explain it than explain it myself. (R)
23.	I'm comfortable being the person who answers the group's questions.
24.	I'd rather attempt problems alone first and bring questions to the group. (R)

Kept identical across all three versions. MSLQ is a study-alone instrument and has no items measuring how someone behaves inside a group, so there is nothing to source here.